

Photograph Utsav Goswami

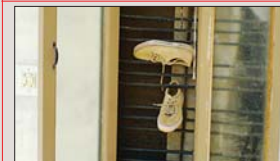
Each month *Digit* will carry a caption for a photo. Come up with something funnier, and beat the *Digit* team at their own game! Entries accepted by the 20th of this month.

Beat That!

"Sania—nice moustache!"

Last Month's Winner!

Dr. Y. Katheeja,
72-SALAI STREET, VEDASANDUR-
624710. "Windows Bata!"



E-mail your caption with the subject "Beat That", and your postal address, to **beatthat@thinkdigit.com** and win

Microsoft Office Access 2007

by Kenneth Hess



Published by

WILEY-INDIA

Reprint, Reuse, Recycle

Some of the most significant computing technologies have originated at the Palo Alto Research Centre (PARC). PARC is a wholly-owned Xerox subsidiary, where research programs are undertaken in areas ranging from biomedical science to ubiquitous computing.

PARC has now researched and developed new printing technologies, one of them being solid ink. This is a wax-based ink in the form of cubes to be used in printers in place of fluid inks. A solid ink stick can print on both sides of a paper simultaneously. Using any kind of media—from photo paper to letterheads—better colour quality can be obtained, too.

Researchers at PARC worked with customers to find that 40 per cent of

Time and research will need to be put into “going green” on a global scale, and it is heartening to see that the major players in the computing space are not only sitting up and taking note, they’re also spreading the word.

BEYOND 80

More Cores

September 2006, Intel unveiled its 80-core teraflop chip prototype. It was part of a research program, and not for commercial purposes. The chip did not contain x86 cores, and could only compute floating points. Other companies and labs have also developed chips with large numbers of cores: ClearSpeed Technology, for example, developed a chip with 96 cores for performing specific tasks.

After having officially showcased the 80-core prototype, questions were shot at Intel: would they make a multi-core chip with x86 cores? Would the

latency, heat, and data traffic problems be solved? Intel merely said they were working upon it.

Now Jerry Bautista, co-director of Intel's Tera-Scale Computing Research Program has said, "Researchers at Intel are working on ways to mask the intricate functionality of massive multi-core chips to make it easier for computer makers and software developers to adapt to them. These multi-core chips will also likely contain both x86 processing cores, similar to the brains inside the vast majority of Intel's server and PC chips today, as well as other types of cores. A 64-core chip, for instance, might contain 42 x86 cores, 18 accelerators, and four embedded graphics cores."

Each core on a massively multi-core chip is connected to an external memory chip through thousands of microscopic interconnects. This technology of connecting chips is known as Through Silicon Vias (TSV).

One idea from a paper released at the Programming Language Design and

Implementation Conference in San Diego has proposed placing all the cores in an “exoskeleton”(the external protective structure/shell like on outer body of insects or other invertebrate) in a heterogeneous multi-core chip, either to make all the cores look like a single core or to make them appear as a series of x86 cores. Another paper presented at the International Symposium on Computer Architecture, also in San Diego, suggests a hardware scheduler that would share computing tasks among the cores. The scheduler would not only speed up processing, it would also prevent heating by transferring tasks to the next non-busy core. Heat issues are also being addressed.

Intel might not have an x86-core-based massively multi-core chip now, but they might well come up with one. That chip we're looking forward to could make Desktop computers efficient enough to emulate supercomputers, which handle computing tasks differently.